Lenses and image formation

antocciphysics | TEACHER KEY | CPA

Learning target: Use ray directions to explain converging and diverging lens behavior.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 11; 3 course days later.*
Observed frequency is higher than 500 Hz because arriving wavefronts are closer together in time.

02. Explain a worked example (worked)
Real inverted image between f and 2 f on opposite side, smaller than the object.

03. Look and explain (model)
Converging rays toward far focus; diverging rays spread with back extensions to near focus.

04. Build the solution (practice)
Inverted real image between f and 2 f for ideal converging lens.

05. Use the idea (reason)
Image character depends object location; inside f emerging rays diverge and back extensions form virtual image.

06. Choose a method (mixed) *Review: Unit 6, Day 2; 12 course days later.*
6 cycles; frequency 1/4 Hz ≈ 0.25 Hz.

07. Exit check (exit)
Diverging.

08. STRETCH: Apply it another way (stretch)
They diverge; backward extensions give an upright magnified virtual image on the object's side.

Lenses and image formation

antocciphysics | TEACHER KEY | HONORS

Learning target: Use ray directions to explain converging and diverging lens behavior.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 11; 3 course days later.*
f_{obs}≈523.1 Hz, higher than emitted frequency.

02. Explain a worked example (worked)
Real inverted image between f and 2 f on opposite side, smaller than the object.

03. Represent and interpret (model)
Converging rays toward far focus; diverging rays spread with back extensions to near focus.

04. Solve and justify (practice)
d_i 15 cm; m=-.5, inverted half size.

05. Analyze the reasoning (reason)
Image character depends object location; inside f emerging rays diverge and back extensions form virtual image.

06. Choose a method (mixed) *Review: Unit 6, Day 2; 12 course days later.*
λ=0.4 m; f=1/T≈16.67 Hz.

07. Exit check (exit)
d_i=-10 cm; m=+2; upright virtual enlarged image.

08. STRETCH: Apply it another way (stretch)
d_i=15 cm; m=-0.5, real inverted half-size image.